Mathematics for Computing - Short Notes

Mathematics for Computing (4COSC007C) · matrices, tableau method and statistics · typed up from handwritten notes and lecture photos

1. Matrices

Addition and subtraction

Two matrices can be added or subtracted only if they have the same order (same rows and columns). Add or subtract the matching elements. e.g. a 2x3 matrix and a 3x2 matrix cannot be added.

Inverse of a 2x2 matrix

A 2x2 matrix A = [[a, b], [c, d]] is invertible if its determinant is non-zero: det(A) = |A| = ad - bc ≠ 0. Then:

```
A^-1 = 1 / (ad - bc) * |  d  -b |
                       | -c   a |
```

Example: A = [[-1, 2], [3, 5]]: |A| = (-1)(5) - (2)(3) = -11 ≠ 0, so A^{-1} exists.

```
A^-1 = 1/-11 * |  5  -2 |  =  | -5/11   2/11 |
               | -3  -1 |     |  3/11   1/11 |
```

Homework: check that $A.A^{-1} = A^{-1}.A = I$.

Inverse of a 3x3 matrix

If A is invertible (|A| ≠ 0, so A^{-1} exists), follow these steps:

1. Find the **Minor** matrix (matrix of minors). M_{ij} is the determinant of the 2x2 matrix left after deleting row i and column j.
2. Find the **Cofactor** matrix: $C_{ij} = (-1)^{i+j} M_{ij}$, i.e. apply the sign pattern [[+, -, +], [-, +, -], [+, -, +]].
3. Find the **Adjoint** matrix: Adj(A) = transpose of the cofactor matrix = C^T.
4. Find the **Inverse**: A^{-1} = (1 / |A|) Adj(A).

Determinant of a 3x3 matrix (expanding along row 1)

```
|A| = a11 M11 - a12 M12 + a13 M13  =  a11 C11 + a12 C12 + a13 C13
    = a11 |a22 a23| - a12 |a21 a23| + a13 |a21 a22|
          |a32 a33|       |a31 a33|       |a31 a32|
```

Any row or column can be used. e.g. expanding along column 2: |A| = -a12 M12 + a22 M22 - a32 M32 = a12 C12 + a22 C22 + a32 C32.

2. Logic - Tableau Method (CPL)

- Let G be the initial graph representing a tableau and G' a reduced graph. G' is empty if the initial node of G has been deleted.
- A tableau is called **closed** if its reduced graph is empty.
- **Statement 1:** for any CPL formula A, a tableau is closed if and only if A is unsatisfiable.
- **Statement 2** (correctness of tableau for CPL): a tableau constructed for the assumption F[A] is closed if and only if A is valid.

Worked example: is ¬(p ∧ q) → (¬p ∨ ¬q) valid?

```
1.  F[ ~(p ^ q) -> (~p v ~q) ]        assume the formula is false
2.  T[ ~(p ^ q) ],  F[ ~p v ~q ]      F of an implication
```

```
3.  F[ p ^ q ],     F[ ~p v ~q ]      T of a negation
4.  F[ p ^ q ],     F[~p], F[~q]      F of a disjunction
5.  F[ p ^ q ],     T[p],  F[~q]
6.  F[ p ^ q ],     T[p],  T[q]
6.1 F[p], T[p], T[q]   -> p = F and p = T: contradiction, branch deleted
6.2 F[q], T[p], T[q]   -> q = F and q = T: contradiction, branch deleted
```

All branches are deleted, so the tableau is closed and the formula is **valid**.

3. Statistics

- **Mode:** the most frequently occurring value.

Measures of spread (ratio data)

1. **Range** = largest - smallest
2. **Mean deviation** (average deviation) = $\Sigma|x_i - \bar{x}| / n$
3. **Variance** $s^2 = \Sigma(x_i - \bar{x})^2 / (N - 1)$
4. **Standard deviation** $s = \sqrt{[\ \Sigma(x_i - \bar{x})^2 / (N - 1)\ ]}$
5. **Coefficient of variation** $CV = s / \bar{x}$
6. **Interquartile range**: Q1 = ¼(n+1)th item, Q2 = ½(n+1)th item (median), Q3 = ¾(n+1)th item; IQR = Q3 - Q1